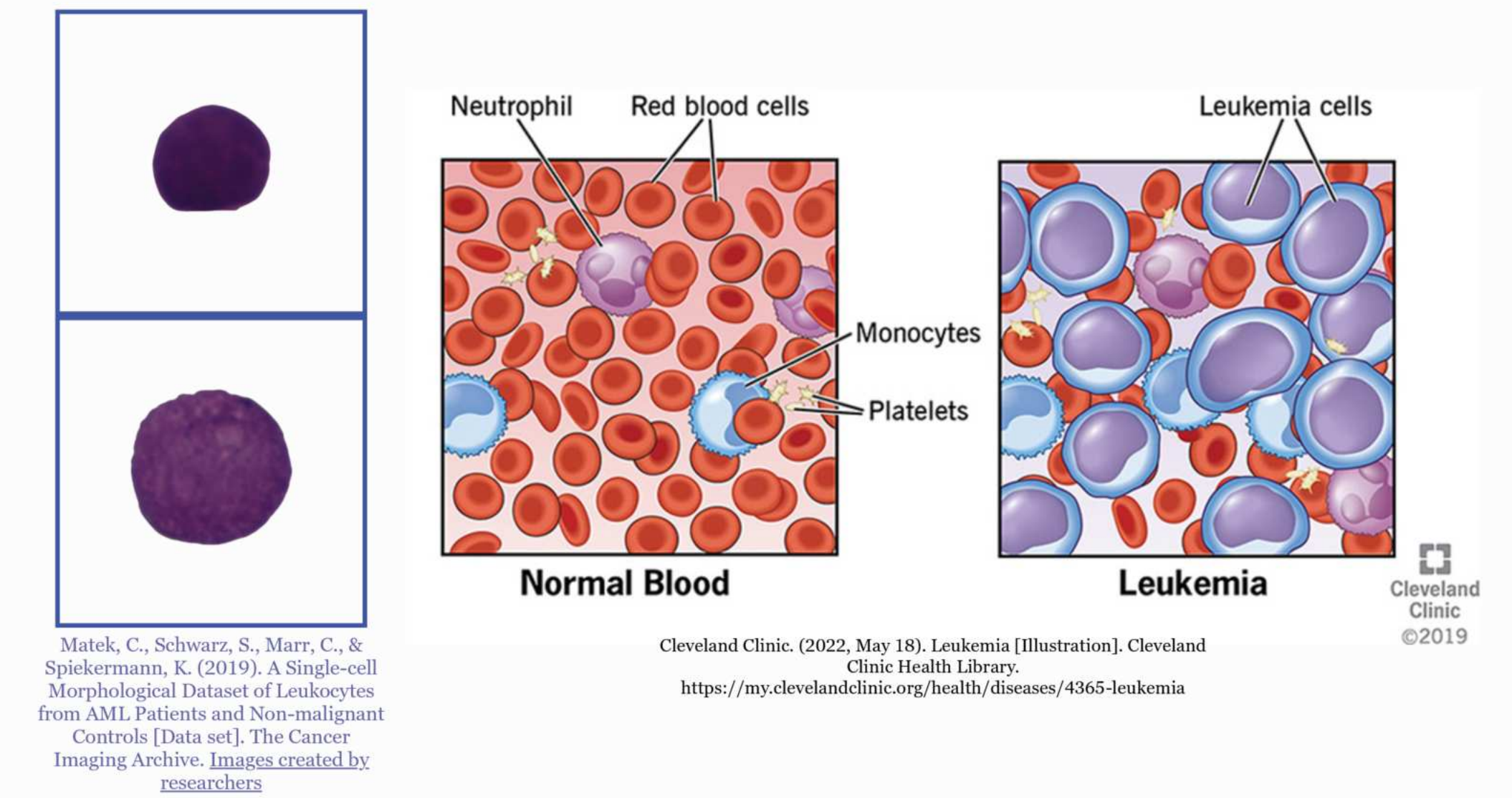


Introduction

Leukemia is a fast-growing cancer of the blood and bone marrow signified by the rapid and abnormal production of immature white blood cells, or **lymphoblasts**. Because these cells multiply rapidly and crowd out healthy blood cells, **early and accurate detection** is crucial for patient survival, especially in children, where Acute Lymphoblastic Leukemia (ALL) is most prevalent. Leukemia represents a significant global health burden with **474,519 new cases** and **311,594 deaths** reported in 2020 alone. Among all types of leukemia, ALL is the most common subtype, accounting for approximately 63% of all cases. Complete Blood Count (CBC) tests are the current method of diagnosis of leukemia through the manual analysis of peripheral blood smears. Digitization of microscope slides can be done by Whole Slide Imaging(WSI) scanners and have shown promise in aiding analysis.



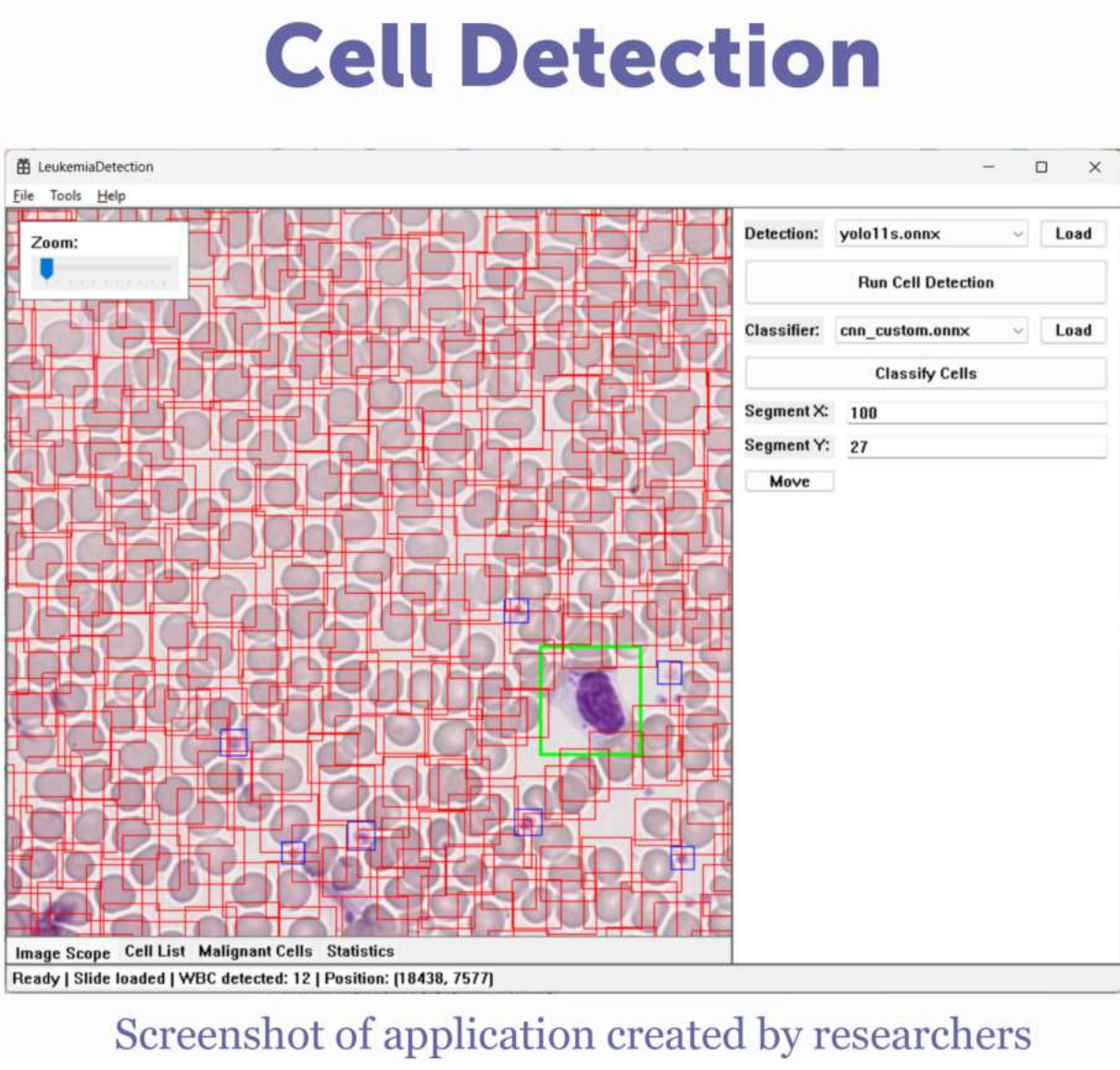
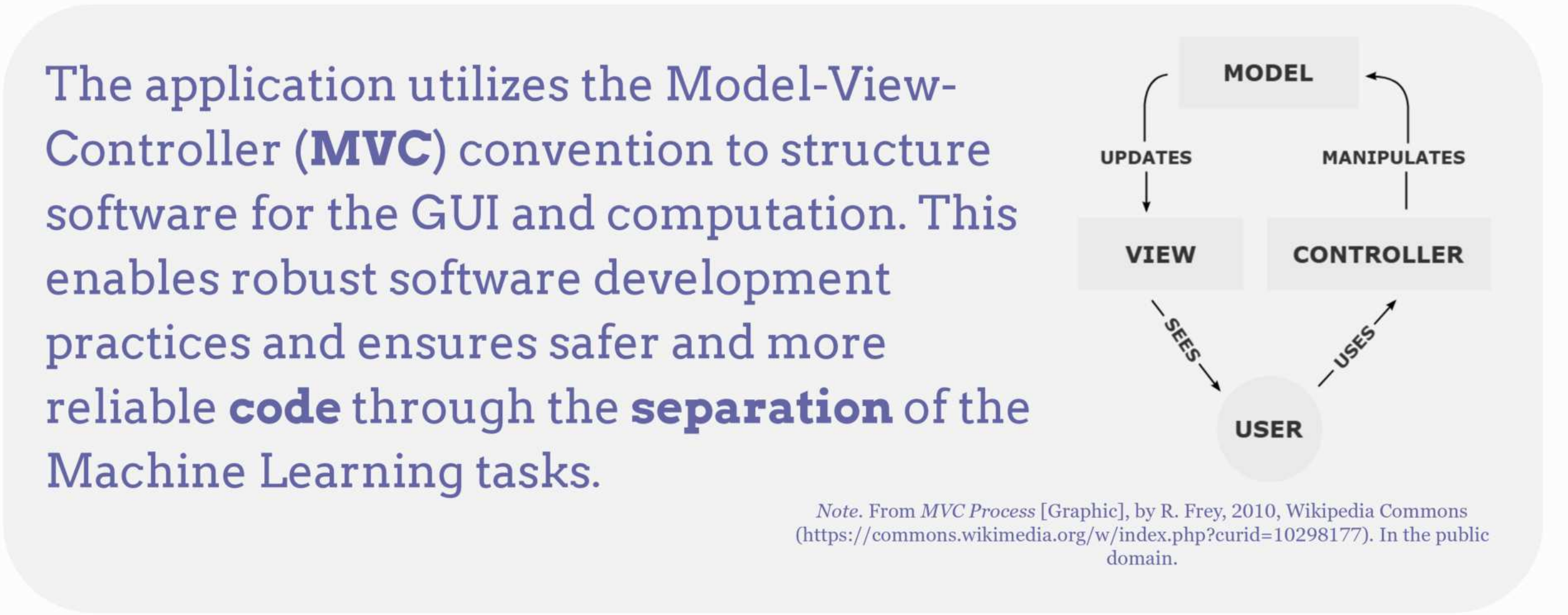
Purpose

While cure rates for children can exceed 85%, the diagnostic process remains a significant bottleneck due to its reliance on the labor-intensive manual analysis of peripheral blood smears by trained lab technicians. In fact, CBC tests in the US, a highly developed country, can reach up to \$100 uninsured. In low-income countries, high mortality rates, some reaching as low as **20%** 5-year survival, are largely due to the lack of access to diagnosis, while 5-year survival rates in developed countries reach **90%**. In recent years, digital pathology has become more accessible, with the mean price for slide scanners dropping from over \$300,000 to under \$1000 in the last decade. We believe that digital aides could help close the gap in diagnostic capability regardless of geographic location.

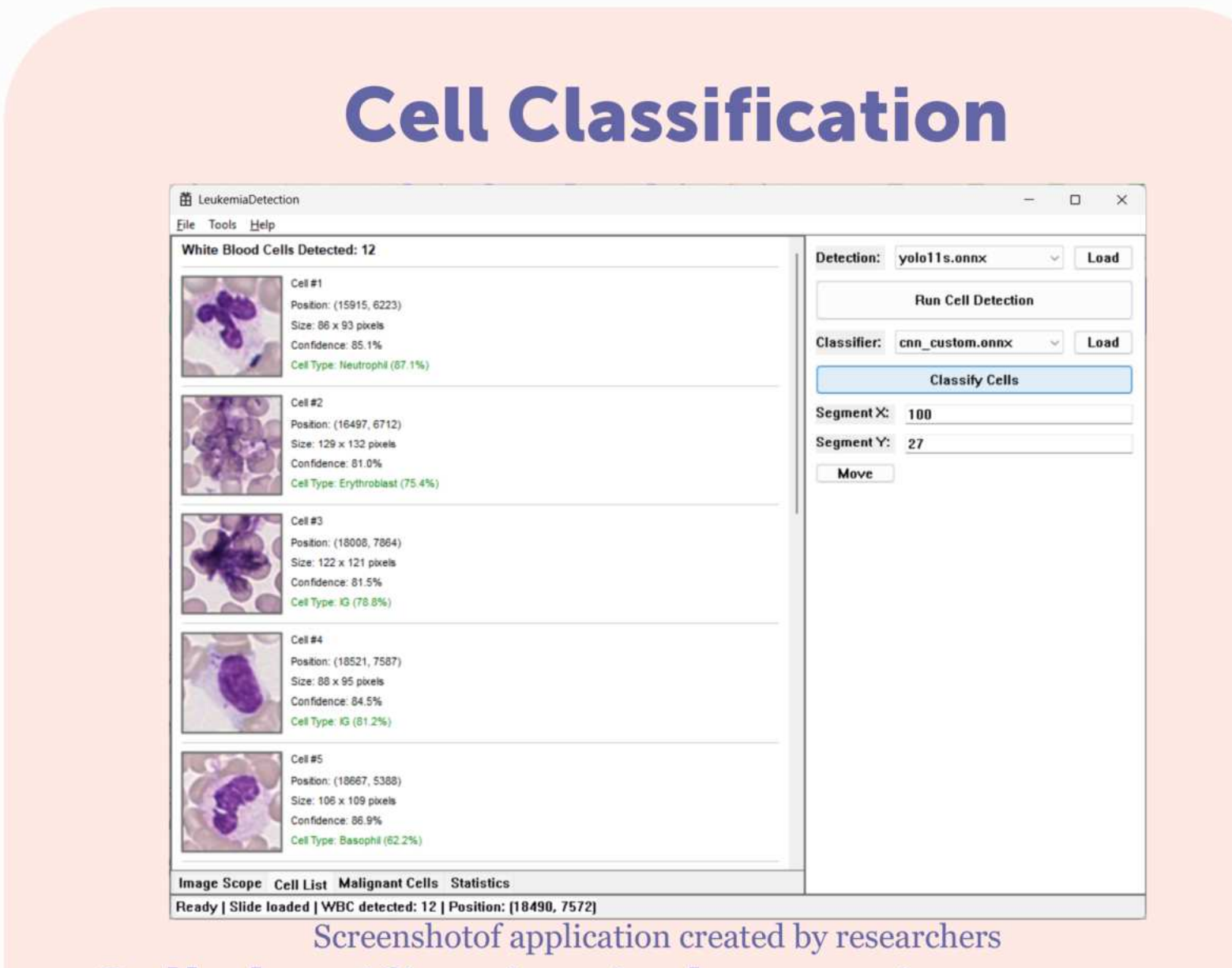
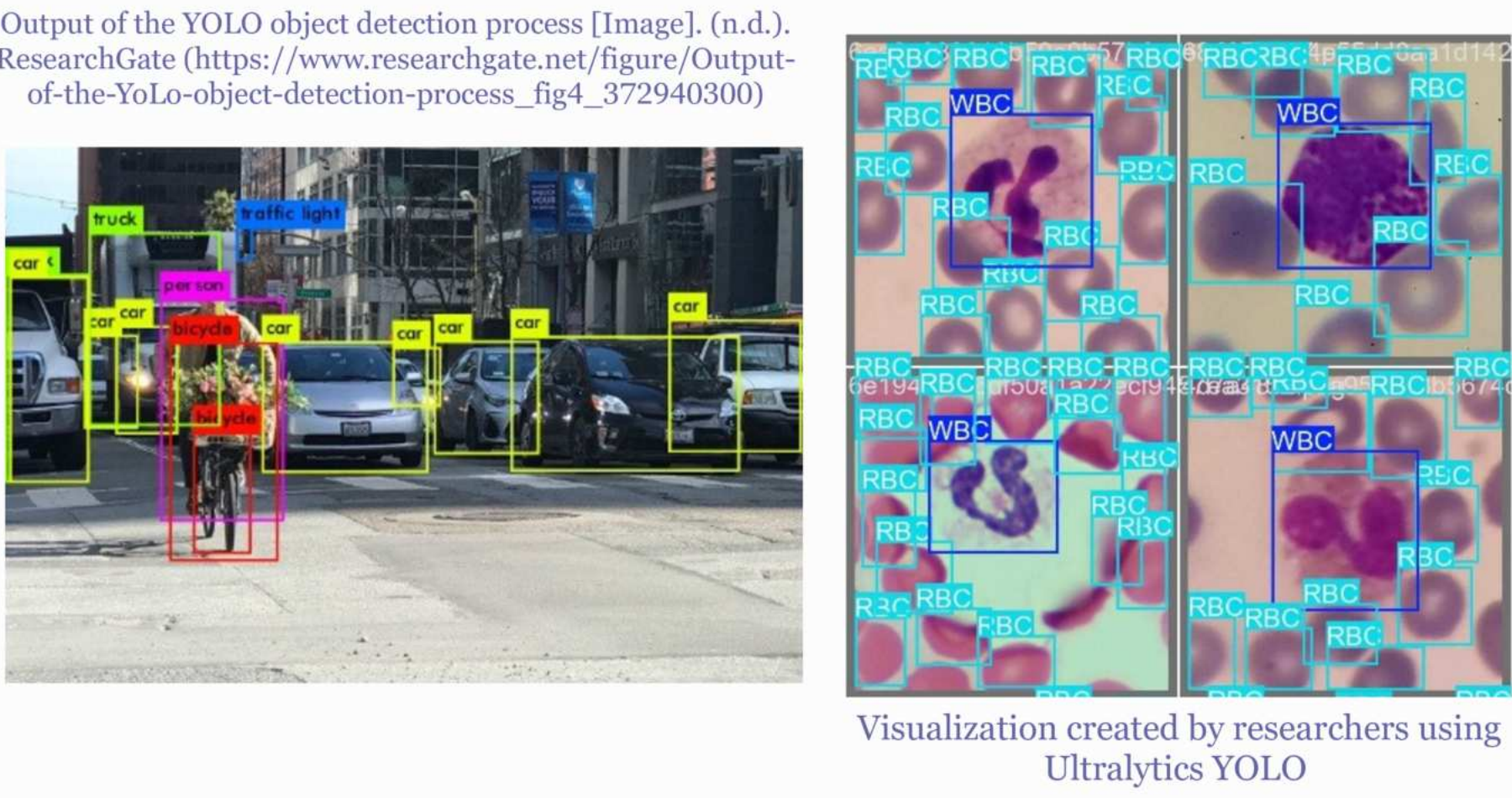
Leukemia Pathology Tools

Our Solution

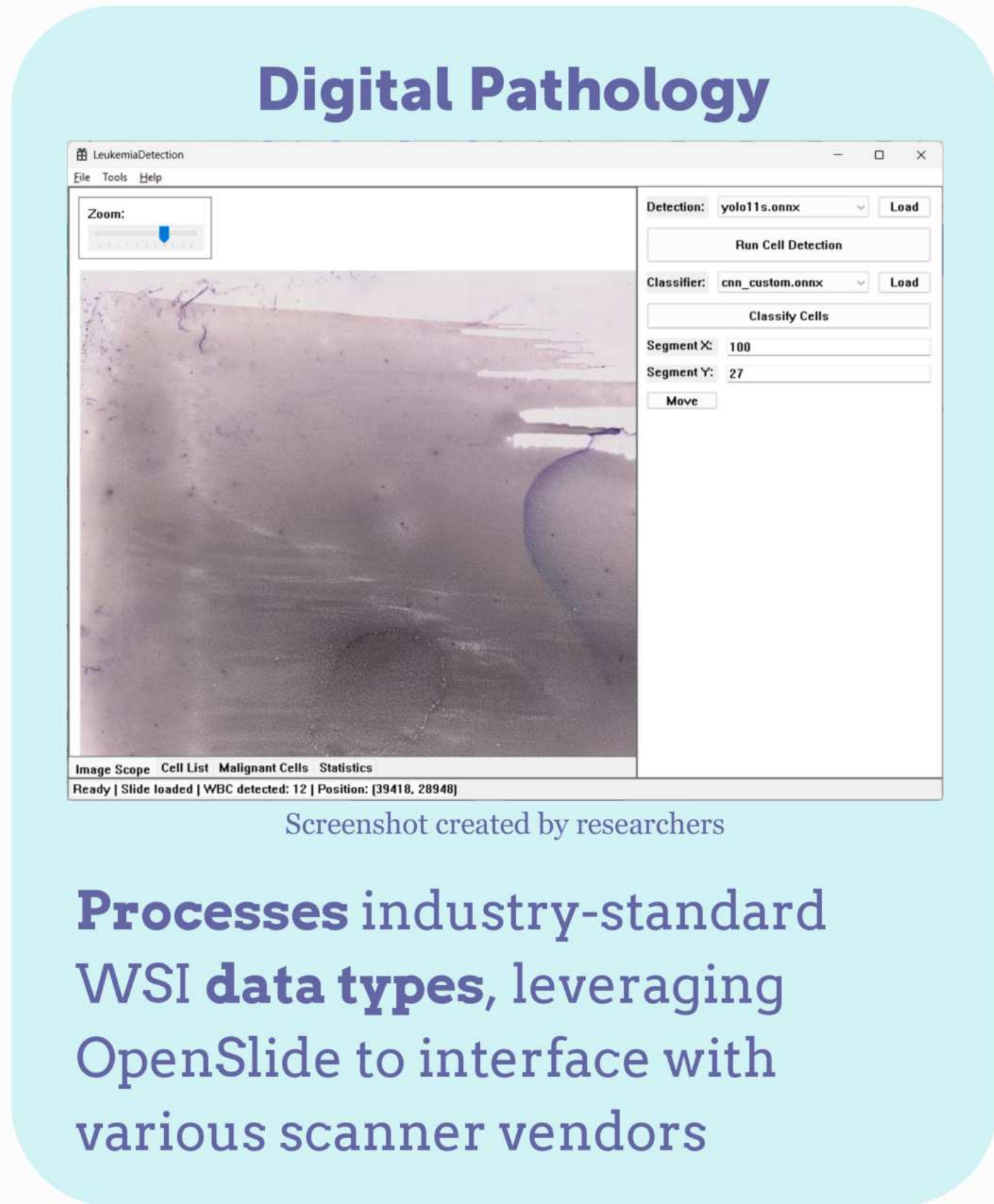
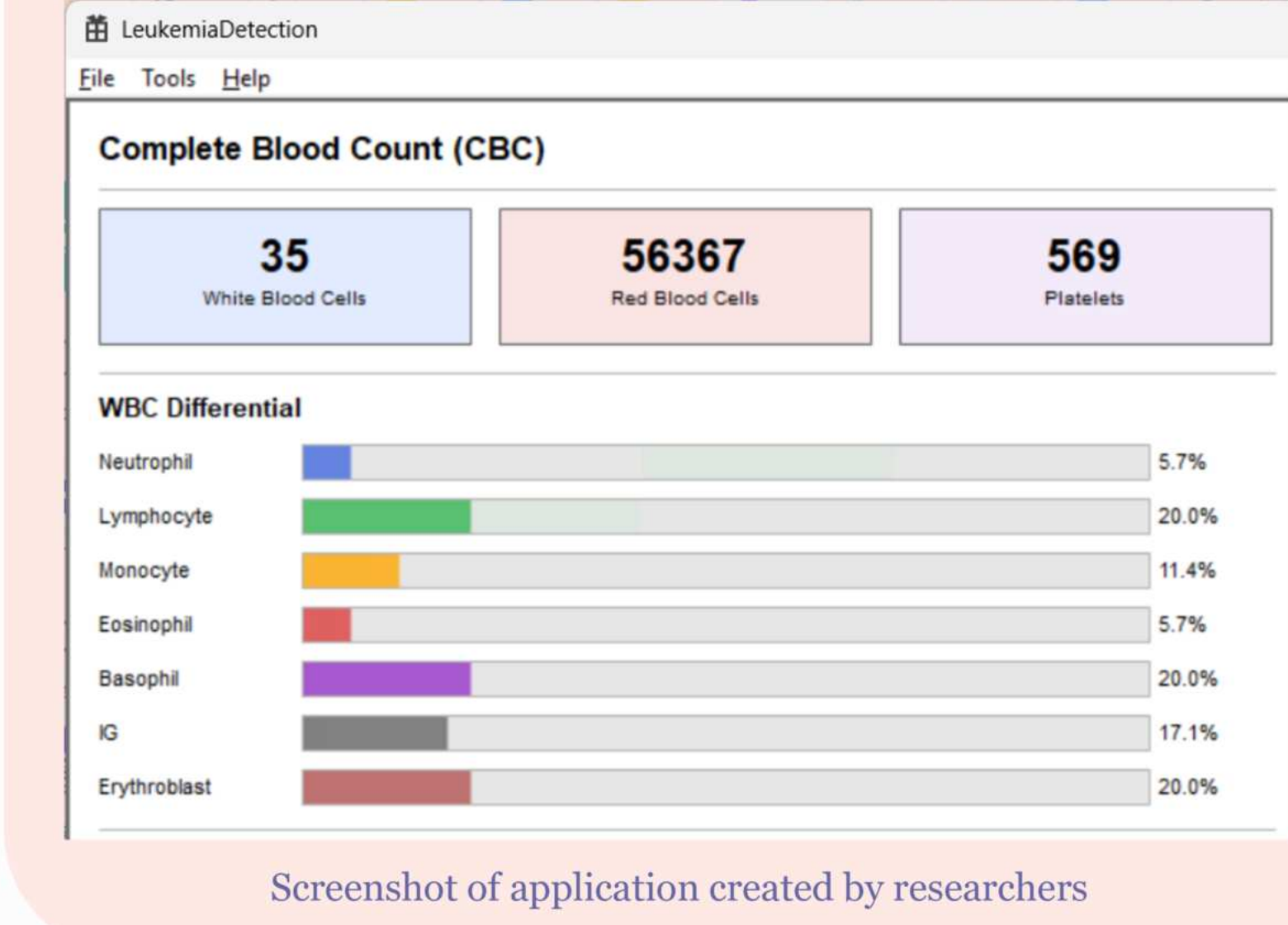
Our approach to the problem is a **Windows desktop application** able to analyze whole slide scans with the aid of **Machine Learning models**.



YOLOv11 is an object detection model optimized for real-time applications. Trained to **detect blood cells**, this model isolates individual cells for subsequent **high-resolution** analysis.



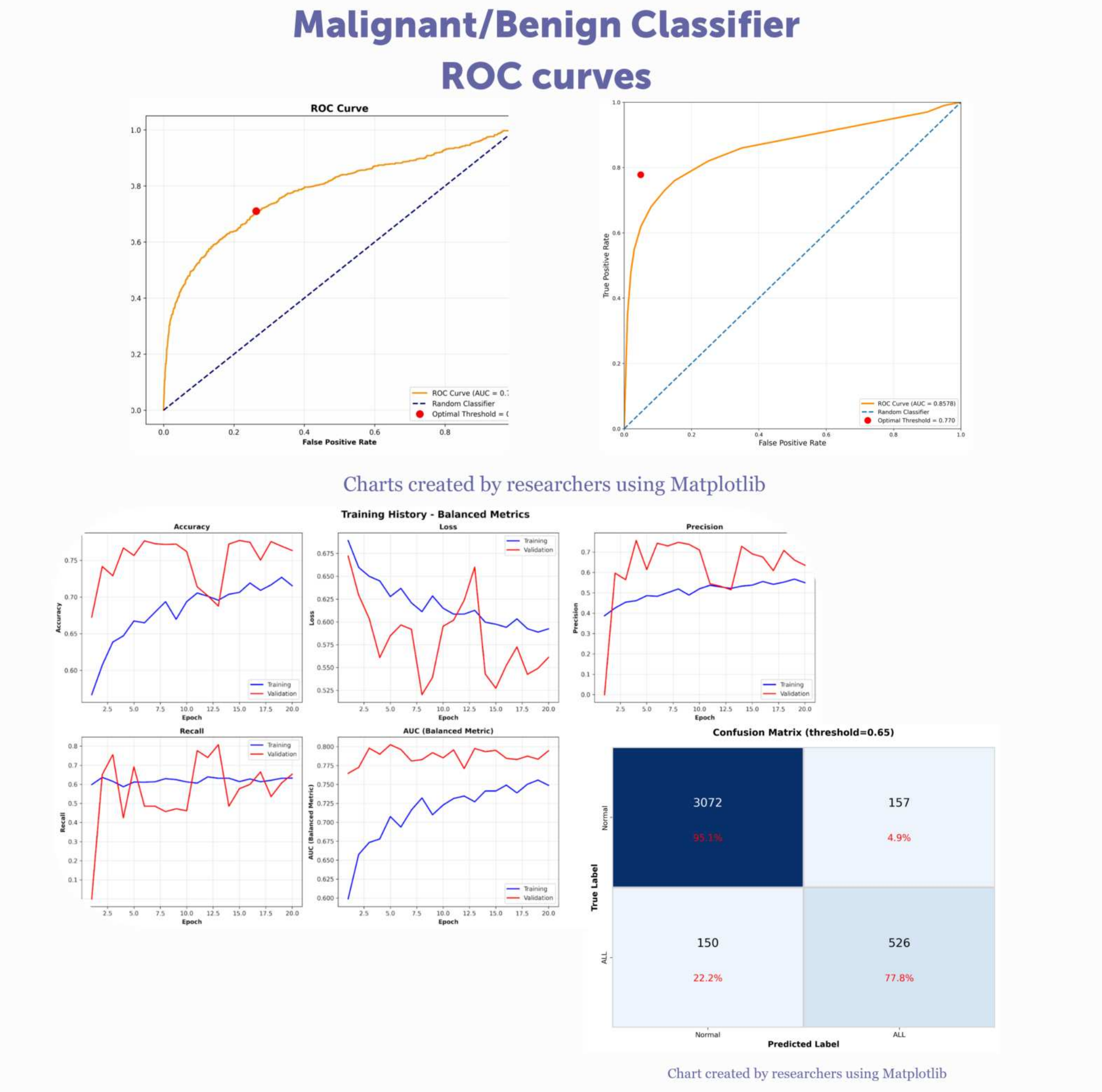
Cell classification is done using a **custom CNN** architecture. The model targets protrusions and distortions, identifying **markers**. The application presents results in various understandable formats.



Processes industry-standard WSI **data types**, leveraging OpenSlide to interface with various scanner vendors

Results

Model Performance Metrics:
Yolo11s mAP-50 - **98.9%**
Yolo11l mAP-50 - **99.2%**
Cell classifier accuracy - **92.3%**



Conclusions

By combining high-speed object detection with deep residual classification in a **native C++** environment, we have created a tool that is both **powerful and accessible**. The ability to run this software offline on widely-available hardware makes it a viable solution for clinics **anywhere in the world** with limited resources.

Next Steps

The next steps for this research include refining the Malignant/Benign Classifier by training on a **larger, more diverse dataset** of pathology images to further increase confidence levels. Additionally, future work will focus on utilizing **LIME**, a machine learning framework used to explain the predictions of machine learning models as well as other explainability features to provide more informed diagnostic tools.

